Predictive Power of Metal Returns on Equity Markets: Evidence from Price Shocks

Introduction & literature

Metals have played a foundational role in human development, initially valued for their industrial utility and later recognized for their intrinsic and financial significance. In modern markets, industrial and precious metals are not only driven by supply-demand fundamentals but also increasingly reflect macroeconomic sentiment, inflation expectations, and investor positioning. This dual role positions metal prices as potential leading indicators of broader economic and financial conditions.

This paper investigates whether industrial metal commodity shocks can predict short-run equity returns of the firms that produce these commodities. Specifically, we ask: Can this predictive power be systematically harnessed to construct profitable investment strategies? The motivation for this research lies in the growing financialization of commodities and the search for alternative macro-sensitive predictors in equity return models. Understanding this relationship offers value to investors seeking abnormal returns and to policymakers monitoring industrial cyclicality.

A growing body of literature supports the economic relevance of metals in financial markets. Jacobsen et al. (2018) argue that metal prices contain predictive information about equity returns, especially in different macroeconomic conditions — positive metal returns are good news in recessions but bad news in expansions. Barsky and Kilian (2002, 2004) highlight the importance of commodity price shocks in reflecting the phases of the business cycle, while Henkel et al. (2011) claim that market-traded, high-frequency indicators, such as metal prices, are particularly well-suited for return forecasting. This study builds on the work of Jacobsen et al. (2018), incorporating their set of industrial metals and expanding the analysis to include zinc. Additionally, we extend the research by forming an investment portfolio based on the predictive power, which in our case primarily focuses on the shock periods.

Given their widespread use in industrial processes, metals such as copper, aluminium, zinc, and nickel are not only economically significant but also highly sensitive to shifts in global growth expectations (Ponomareva et al., 2024). These characteristics make them particularly promising candidates for testing predictive return dynamics. In addition, oil is included as a benchmark, due to its well-documented role in influencing equity markets and its frequent use in related literature as a macro-financial variable (e.g., Kilian & Park, 2009).

The remainder of the paper is structured in the following manner. The data section describes the data sources that were used and provides a general overview of the retrieved data. The Analysis & Results section details how the data was used to find the predictive power of the industrial metal returns on the future equity returns. Thereafter, an investment strategy is designed to harness this predictive power. The paper finishes with the Conclusion & Discussion, providing the implications and explanations of the results and provides possible limitations of the model.

Data

To estimate the linear regression models and construct commodity-specific investment strategies, the Federal Reserve Economic Data (FRED) database was used to retrieve monthly US data on commodity prices, as well as the inflation rates and term spread control variables. Monthly equity price data corresponding to each commodity-related US firm was collected from the Center for Research in Security Prices (CRSP). The resulting database constitutes a panel comprising 420 observations over the period from 1990 to 2024.

The data encompasses 582 unique equities across the 5 commodities, although the number of equities is mainly centered in oil with 510 firms. Moreover, the average market cap is relatively small across all metals and oil, but all these firms have a high average listing period on the market (Table 1). Table 2 and Figure 1 show us the equity return statistics and the return correlation for each commodity respectively. Particularly, Table 2 shows that Aluminium has by far the lowest mean monthly returns, while Copper has the best mean monthly returns. Moreover, Zinc and Nickel have the highest annualized volatility. Figure 1 presents a moderate correlation between metal returns and a low correlation with the benchmark commodity, Oil.

All commodity and equity prices were resampled to a monthly frequency, and returns were calculated accordingly. Using shares outstanding and market price data, market capitalizations were computed to construct the value-weighted portfolios.

This dataset presents several advantages. The extensive timeframe facilitates the examination of long-term dynamics including multiple economic cycles and commodity price shocks. Additionally, the cross-sectional variation across firms enhances the robustness of the results. A limitation of our dataset is the unequal representation of equities across commodities, with oil having 510 equities compared to only 11 for zinc. This imbalance could influence the relative statistical power across commodity-specific regressions. Moreover, the equities are all US-based which makes it difficult to extrapolate our findings to a global scale.

Analysis & results

For the first part of the analysis, we examine the predictive power of the described metals on the next month's equity returns of the firms that produce the metals. To this extent, we run a regression analysis, where we distinguish between 3 different types of periods for all industrial metals. Namely, the periods where no price shocks of the metals, positive price shocks and negative price shocks take place. This is in line with the findings of oil markets, where there seems to be a difference in predictive power between no-shock and shock periods. Moreover, it is shown that there exist an asymmetric effect between negative and positive shock periods (Gazzani et al., 2024). In all the regressions the dependent variable will be the weighted returns of the producing companies of the specified metals ($r_{i,t}^{industry}$) and the independent variables will be the metal returns of the previous periods ($r_{i,t-1}^{metal}$) for all the industrial metals, where i represents the specific metal

and t the time period. This will provide us with the effects of the returns of the industrial metals and oil on its specific producing industry. The complete model with all variables can be observed in the Predictive power model.

$$r_{i,t}^{industry} = \beta_0 + \beta_1(r_{i,t-1}^{metal} * D_{i,t-1}^{noShock}) + \beta_2(r_{i,t-1}^{metal} * D_{i,t-1}^{pos}) + \beta_3(r_{i,t-1}^{metal} * D_{i,t-1}^{neg}) + \gamma_i Control_{i,t-1} + \epsilon_t \quad \text{(Predictive power model)}$$

where the following elements are defined:

1. The Control variables incorporate inflation and term spreads (10Y - 2Y).
2. The metals that are used for this analysis are: Copper, Aluminium, Zinc and Nickel. Moreover, Oil is used as a benchmark.
3. $D_{i,t-1}^{pos}$ represents a dummy variable in case of a positive price shock, where $D_{i,t-1}^{\text{pos}} = \mathbb{I}\left(r_{i,t-1}^{\text{metal}} \geq 2\hat{\sigma}_{12}(r_{i,t-1}^{\text{metal}})\right)$ where $\mathbb{I}(\cdot)$ is the indicator function and $\hat{\sigma}_{12}$ denotes the 12-month rolling standard deviation.
4. $D_{i,t-1}^{neg}$ represents a dummy variable in case of a negative price shock, where $D_{i,t-1}^{\text{neg}} = \mathbb{I}\left(r_{i,t-1}^{\text{metal}} \leq -2\hat{\sigma}_{12}(r_{i,t-1}^{\text{metal}})\right)$ where $\mathbb{I}(\cdot)$ is the indicator function and $\hat{\sigma}_{12}$ denotes the 12-month rolling standard deviation.
5. $D_{i,t-1}^{noShock}$ represents a dummy variable in case of no price shock periods, where $D_{i,t-1}^{noShock} = 0$ if there is a shock in the returns of the metal ($D_{i,t-1}^{pos} = 1$ or $D_{i,t-1}^{neg} = 1$) and 1 otherwise

The definition used for price shocks builds upon established methodologies of papers that examined shocks in the markets that are also examined in this research paper. Specifically, the frameworks that are used in this study employ similar methods from studies on the predictive power of oil shocks on equity markets (Hammoudeh et al., 2003) and return shocks in equity markets (Govindaraj et al., 2013). This definition will ensure that this study is consistent with existing literature. To account for the macro economic situation of the USA, we use inflation and term spread (10Y - 2Y) as control variables (Jacobsen et al., 2018).

Table 3 and 4 presents the regression results of the predictive power of the model for all the industrial metals using equal-weighted industry returns (Table 3) and value weighted industry returns (Table 4). From table 3 and 4, it can be noted that the negative shocks are for some metals statistically significant in predicting returns on producers of the commodities in the next month, while the positive shock and no shock case show no significance for any of the commodities. This suggests that sharp price declines of the commodities exhibit significant predictive power on the producer's returns of the next month for some commodities, while neutral or positive movements do not.

A reason the predictive power in Tables 3 and 4 prove to be insignificant, is due to the price change effect occurring in the same month, rather than the next month. In Tables 5 and 6, it is observed that negative shocks and no-shock periods of commodity returns have a significant effect on equity returns in the same

month, providing a possible explanation of the results in Tables 3 and 4. This implies that the price of the stocks mainly changes in the same month and not the month after, which is in line with the efficient market hypothesis. With negative shocks of some commodities still having significant predictive power on the producers' returns in the next month, we will focus our investment strategy on these shocks. Although positive shocks do not hold significant predictive power in our model, they will be included in our strategy, as in literature positive shocks have proven to also hold predictive power (Knop & Vespignani, 2014).

To take advantage of this predictive power of price shocks, we form a 'composite long-short' strategy. This entails investing in the S&P 500 during periods with no shocks in any of the industrial metals that we examined. We follow the S&P 500 in case of no shock periods, as most of the time there are no shocks in the industrial metals and in such cases we still want to have the money invested. Additionally, the S&P 500 is a widely used metric to form a proxy of the market portfolio. However, during periods of price shocks of a specific metal, we will sell our position in the S&P 500. After selling this position, we go long on the equal-weighted equity portfolio of the industry that produces this metal and in case of a negative shock we will short the equity portfolio of the industry that produce the metal. The results of this strategy are shown in Figure 2, see Appendix D.

As can be derived from Figure 2, the composite Long-Short strategy yields a cumulative return multiplier of around 70 over a period from January 1990 to January 2025, excluding transaction costs. In comparison to the benchmark, the S&P 500, yields a cumulative return multiplier of around 18 over the same period. Although the strategy seems to be drastically outperforming the benchmark, the risk-adjusted returns of the two strategies are similar, as can be observed in Table 7 in Appendix C. From Table 7 we can observe the relative performance of the two strategies, highlighting that the annualized returns of the composite strategy are somewhat larger, but similarly its annualized volatility is also somewhat larger. This has lead to similar Sharpe ratios of the two strategies. We can also observe that the strategy generates significant α at about 9%. Figure 3 shows the number of turnover events of our strategy over the years, which accumulates to 328 turnover events. This significant number of turnover periods highlights the importance of transaction costs. The transaction costs that were used in this paper are 5 basis points for long positions and 10 basis points for short positions. As can be observed from Figure 2, the strategy still performs really well with a cumulative return multiplier of around 60. From table 7 we can observe that the Sharpe ratio remains slightly higher than the S&P 500 strategy.

Lastly, from Figure 4, we can observe that for the industrial metals we maintain relatively more long positions compared to short positions, while the opposite is true for the oil positions. Moreover, it can be observed that we maintain a similar amount of positions between all the industrial metals and oil over the investment period.

Conclusion & Discussion

The first objective this study set out to find is whether industrial metals, in particular copper, aluminium, zinc, nickel, and oil can predict future returns of the US companies that produce them. To this extent, we run regressions using both an equal and value weighted industry returns. We found that for most of these industrial metals that negative price shocks can predict short run equity returns of the producing industries, while positive price shocks and no-shock period returns can not. These findings contradict with the results of Knop and Vespignani (2014), where they found that positive shocks were significant predictors. This can be explained by their data being limited to Australia and by their use of other commodities compared to this paper.

This leads to the second objective this paper set out to find, which is whether we can systematically harness this predictive power to construct profitable investment strategies. Using the results of the first part of the analysis, we formulated an equal weighted portfolio based on both positive and negative price shocks. Although positive shocks do not hold significant predictive power in our model, it will be included into our strategy, as in literature positive shocks have proven to also hold predictive power (Knop & Vespignani, 2014). We have shown that our model significantly improves the returns of a strategy that only buys the S&P 500 at the beginning of our time period. However, the additional returns are followed by higher volatility. This has lead to a slightly higher Sharpe ratio of our strategy in comparison to the proxy of the market portfolio. Therefore, our portfolio is designed for investors that are inclined to take more risk for slightly higher risk adjusted returns. Figure 2 provides a possible explanation for the out performance, where especially around crisis periods (e.g. 2008, 2020) the number of turnovers is the highest. Financial crisis also affect the metal prices, leading to shock movements in the metal prices during those periods. This could imply that our strategy might sell its position in the S&P 500 during years of crisis, which is the time where the S&P 500 performs the worst. Therefore, our strategy might take advantage of financial crisis, which is a time with more volatility, explaining why both the returns and volatility of the portfolio are larger. Table 8 contains interest insights into our strategy return contribution by position, where our short-side dominates the long-side whilst the S&P500 contributes the most to our cumulative portfolio returns.

Despite the strong results, there are limitations to our model. The first limitation of the model is the generalizability of the results. This study only incorporated US firms, but future research could test if the results also hold in other economies. The second limitation is the robustness of the result, as for example we did not test different definitions such as for shocks and did not do out of sample validation. However, as we used similar methods as in previous literature, we believe our results are still valid. Lastly, as shown in Table 1, the number of the firms that produce industrial metals is rather limited. This could introduce a possible concentration bias on this limited number of firms. Nonetheless, due to the long time period that was covered, we believe this dependency does not have a strong impact on the results.

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Appendix

Appendix A - Data Description and Summary Statistics

Table 1: Firm-Level Descriptive Statistics by Metal

Metal	Number of Firms	Avg Market Cap (M$)	Avg Listing Length (Years)
Copper	30	3.02	17.72
Aluminium	10	4.28	25.48
Zinc	11	2.86	28.15
Nickel	21	0.64	18.98
Oil	510	3.38	17.62

Table 2: Combined Equity Return Statistics per Commodity

Commodity	Mean Monthly	Std Dev	Min	Max	Ann. Return	Ann. Volatility	Sharpe
Copper	0.0156	0.1043	-0.3161	0.3513	0.1873	0.3614	0.5184
Aluminium	0.0047	0.0999	-0.3600	0.3300	0.0564	0.3461	0.1629
Zinc	0.0116	0.1597	-0.5638	1.7069	0.1388	0.5533	0.2509
Nickel	0.0100	0.1386	-0.4119	0.5925	0.1198	0.4800	0.2497
Oil	0.0112	0.0907	-0.3919	0.5231	0.1347	0.3143	0.4285

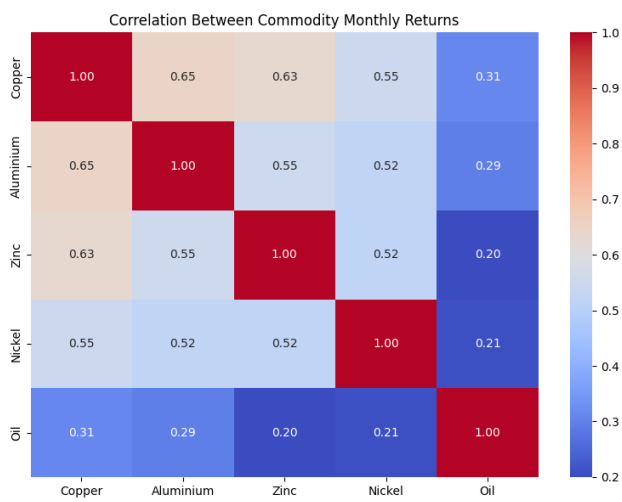

Figure 1: Correlation matrix of commodity returns

Appendix B - Predictive power

Table 3: Equal-Weighted predictive power results by metal

Variable	**Copper**	**Aluminum**	**Zinc**	**Nickel**	**Oil**
constant	0.0195**	0.0078	0.0158	0.0141	-0.0134
	(2.09)	(0.88)	(0.84)	(1.14)	(-1.63)
Positive shock	0.1118	-0.0146	0.1297	-0.2619	0.0688
	(0.51)	(-0.05)	(0.35)	(-1.10)	(0.37)
Negative shock	0.2273	0.5300*	0.6801**	0.2402	-0.2752***
	(1.08)	(1.88)	(1.99)	(0.85)	(-3.06)
No shock	0.0795	0.2025	0.0331	0.1364	-0.0400
	(0.73)	(1.59)	(0.17)	(1.40)	(-0.71)
Inflation	0.0023	-0.0020	-0.0235	-0.0063	0.0818***
	(0.11)	(-0.10)	(-0.62)	(-0.24)	(4.21)
Term Spread	-0.0041	-0.0014	0.0022	-0.0008	0.0046
	(-0.73)	(-0.27)	(0.20)	(-0.11)	(0.96)

Notes: t-statistics are reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$

Table 4: Value-Weighted Predictive Power Results by Metal

Variable	**Copper**	**Aluminum**	**Zinc**	**Nickel**	**Oil**
constant	0.0212**	0.0144	0.0261	0.0228	-0.0202
	(2.37)	(1.50)	(1.29)	(1.39)	(-0.98)
Positive shock	0.1035	-0.2107	0.1709	-0.4210	0.0000
	(0.50)	(-0.70)	(0.47)	(-1.41)	(-1.37)
Negative shock	0.1532	0.7452**	0.7355**	-0.4193	-0.5307***
	(0.77)	(2.47)	(2.12)	(-0.57)	(-3.96)
No shock	0.1357	0.2119	-0.1007	0.0375	0.2194
	(1.31)	(1.56)	(-0.50)	(0.25)	(1.14)
Inflation	0.0072	0.0026	-0.0208	-0.0152	0.0703
	(0.37)	(0.13)	(-0.53)	(-0.43)	(1.35)
Term Spread	-0.0037	-0.0048	-0.0083	-0.0100	-0.0096
	(-0.70)	(-0.84)	(-0.68)	(-0.83)	(-0.80)

Notes: t-statistics are reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Appendix C - Cross sectional regressions

Table 5: Equal-Weighted cross sectional power results by metal

Variable	**Copper**	**Aluminum**	**Zinc**	**Nickel**	**Oil**
constant	0.0276**	0.0127	0.0112	0.0160	0.0073
	(3.18)	(1.50)	(0.63)	(1.34)	(1.10)
Positive shock	0.2981	0.3030	0.6594*	0.0614	0.6072***
	(1.46)	(1.14)	(1.84)	(0.26)	(3.91)
Negative shock	0.6630***	0.4698*	1.0299***	0.6196**	0.5729***
	(3.45)	(1.72)	(3.22)	(2.32)	(7.88)
No shock	0.6872***	0.8430***	0.6099***	0.3247***	0.4358***
	(6.69)	(6.98)	(3.24)	(3.38)	(10.00)
Inflation	-0.0311*	-0.0226	-0.0207	-0.0175	0.0051
	(-1.66)	(-1.23)	(-0.58)	(-0.69)	(0.36)
Term Spread	-0.0057	-0.0034	0.0049	-0.0007	0.0021
	(-1.09)	(-0.67)	(0.46)	(-0.09)	(0.52)

Notes: t-statistics are reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$

Table 6: Value-Weighted Cross-Sectional Power Results by Metal

Variable	**Copper**	**Aluminum**	**Zinc**	**Nickel**	**Oil**
constant	0.0287***	0.0196**	0.0209	0.0258	0.0521***
	(3.46)	(2.14)	(1.09)	(1.62)	(3.03)
Positive shock	0.3486*	0.2749	0.5901*	0.2624	0.0000
	(1.79)	(0.96)	(1.69)	(0.89)	(-3.74)
Negative shock	0.5330***	0.7841***	0.9714***	0.8218	0.4913***
	(2.90)	(2.67)	(3.03)	(1.14)	(4.47)
No shock	0.6810***	0.8460***	0.7246***	0.3687**	0.7204***
	(6.90)	(6.51)	(3.81)	(2.50)	(4.36)
Inflation	-0.0234	-0.0220	-0.0174	-0.0366	-0.1095**
	(-1.31)	(-1.11)	(-0.48)	(-1.04)	(-2.70)
Term Spread	-0.0054	-0.0067	-0.0056	-0.0088	-0.0014
	(-1.07)	(-1.23)	(-0.48)	(-0.75)	(-0.13)

Notes: t-statistics are reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$

Appendix D - Portfolio results

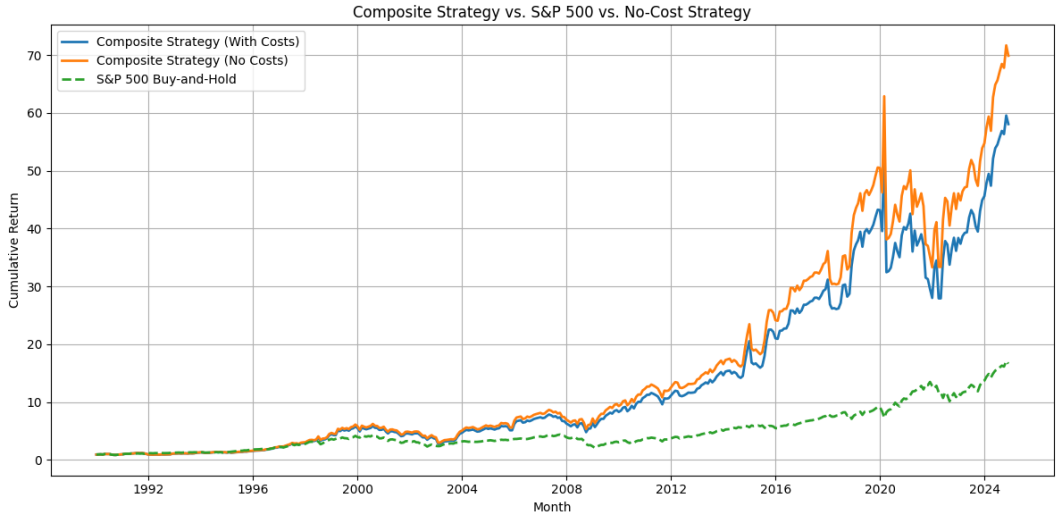

Figure 2: Cumulative returns of the composite strategy (including and excluding transaction cost) and buy and hold S&P 500 strategy

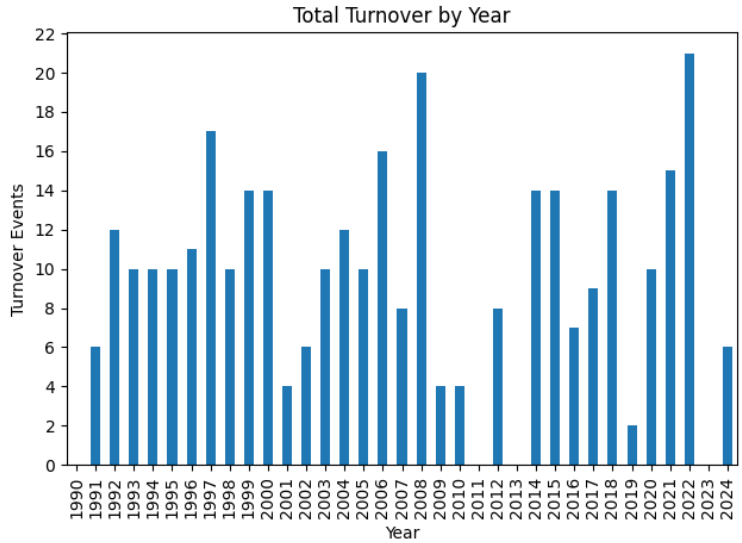

Figure 3: Total number of turnovers (the number of entry and exit events of every month) per year under the composite strategy

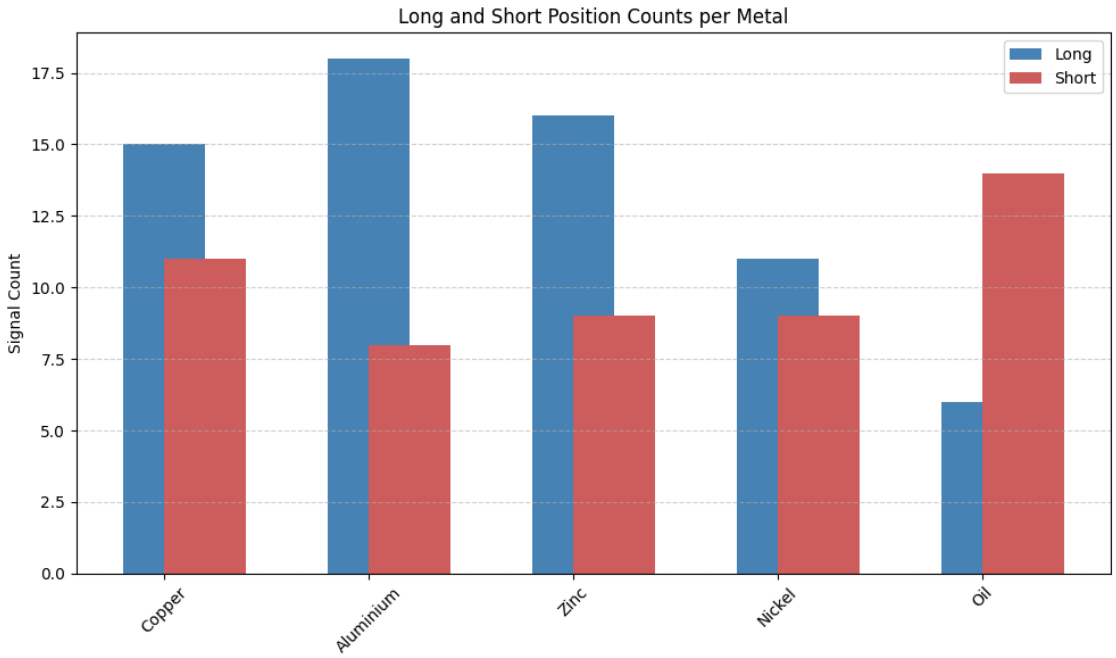

Figure 4: Total number of Long and short positions per metal and oil

Table 7: Performance Comparison of the Strategies

Strategy	**CAGR**	**Annualized Return**	**Annualized Volatility**	**Sharpe Ratio**	**Annualized α**
Composite with costs	0.1263	0.1438	0.2183	0.6588	0.0953
Composite without costs	0.1290	0.1462	0.2185	0.6693	0.0980
S&P 500	0.0837	0.0917	0.1481	0.6189	–

Table 8: Return Contribution by Strategy Component

Component	**Return Contribution**	**Share of Total Return**
Long Positions	0.7190	14.28%
Short Positions	1.9544	38.82%
S&P 500 Fallback	2.3612	46.90%
Total	**5.0346**	**100.00%**